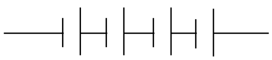
| | | | |
|---|---|---|---|
| 7 | a | Any reasonable suggestions as a battery / power source for pacemakers, RFID, camera and other wearable devices. | A1 |
| | b | Current density is current per unit area.<br>Energy density is energy per unit mass.<br>(Answers in terms of units not accepted) | A1<br>A1 |
| | c(i) | Given charge capacity is 3.6 C $cm^{-2}$,<br>Charge storage capacity = 3.6 x 50<br>= 180 C | <br><br>A1 |
| | (ii) | Given recommended max. discharge current density is 0.15 mA $cm^{-2}$,<br>recommended max. discharge current = 0.15 x 50<br>= 7.5 mA | <br><br>A1 |
| | (iii) | $Q = It$<br>$180 = (7.5 \times 10^{-3})t$<br>$t$ = 24 000 s | <br>C1<br>A1 |
| | (iv) | $E = QV$<br>= (180)(2.5)<br>= 450 J | <br>C1<br>A1 |
| | d(i) | The initial charging current is around 40 mA but decreases rapidly to about 10 mA and stays constant for around 2 hrs before decreasing gradually to zero at the end of the 5 hr charging period. | B1 |
| | (ii) | The area represents the total charge supplied to the cell. | B1 |
| | (iii) | 9 mA. (Acceptable range: 8 – 10 mA) | A1 |
| | (iv) | $E = QV$<br>$= (9 \times 10^{-3})(5 \times 60 \times 60)(3.4)$<br>= 551 J | <br>C1<br>A1 |
| | e | Efficiency = (discharged energy / energy used to charge cell) x 100%<br>= (450 / 551) x 100%<br>= 81.7%<br><br>(Do not accept answer > 100%) | <br>C1<br>A1 |

| | | | |
|---|---|---|---|
| | **f** | Any valid reasons for two of the safety considerations:<br><br>(1) Short circuited results in large currents which cause overheating, cell may catch fire; result in burn hazard.<br><br>(2) Above 140 $^{o}$C, cells may overheat and catch fire; result in burn hazard Or plastic film may melt.<br><br>(3) Water or water vapour within lithium cells may result in electric shocks Or lithium reacts violently with water. | **B2** |
| | **g** | Drawing of 4 cells arranged in series<br>as each cell has average e.m.f. of 2.5 V.<br><br><br><br>Stating that each cell has an electrode area of 2000 cm$^2$<br>based on recommended max current density of 0.15 mA per cm$^2$. | **B1**<br><br><br><br><br>**B1** |
| | | Total: | **20** |

## Examiner's comments for Data Analysis Question (N93/II/8)

*Q.8* Answers: *(c) (i)* 180 C, *(ii)* 7.5 mA, *(iii)* 24 000 s, *(iv)* 450 J

*(d) (i)* 9 mA, *(ii)* 550 J, *(e)* 79%, *(g)* 4 cells in series each of area 2000 $cm^2$

Not many candidates were able to work completely through the whole of this question. Almost any answer was accepted in part *(a)* as a use of the cell, except, as a car battery. Rather too many candidates did not state a use however, they simply wrote 'in a rechargeable battery'. In part *(b)* the only frequent difficulty was with candidates who could not see that the unit watt-hour is a unit of energy. Many thought it was a unit of power and many thought it was a unit of power per hour. In part *(c)* most were able to find the charge-storage capacity and the maximum discharge current but because the output voltage falls during discharge they assumed that the output current must necessarily fall also. If the load resistor's value is changed it is possible to supply a constant current of 7.5 mA for 24 000 s. Some weaker candidates saw the word capacity and tried to make the cell a capacitor. A wide range of answers was accepted for the average current in part *(d) (i)*, between 8 mA and 10 mA. All that was expected was that the candidate would use a (preferably transparent) ruler on the graph in order to have equal areas above and below the ruler. A corresponding range of efficiencies were also allowed but it was extraordinary how many candidates, in order to obtain efficiencies less than 100%, simply put their numbers the wrong way up, rather than using their answer as a check on earlier working. Others were quite happy with efficiences greater than 100% in some cases and ~0.001% in other cases. Part *(f)* was answered sensibly by most candidates but part *(g)* caused unexpected difficulty. A surprising number of candidates did not even realise that four cells in series would be needed for 10 V. Many also assumed that 500 $cm^2$ for each of the four cells would be sufficient. Those who assumed that the cells were still the ones of 50 $cm^2$ did have problems attempting to draw the circuit.